

Unnamed_Unit

Unit 1 - If, If Else

PDF Documents

PDF Document 1: 1738153849701-Chapter 10 - Unit 1 - If, If Else.pdf

This PDF document is included in the unit materials.



UNIT 1 IF, IF ELSE

If Statement

WHAT IS AN 'IF' STATEMENT?

An "if statement" in programming is like a decision-making step that helps a computer decide what to do based on whether a condition is true or not. It's like asking a question, and if the answer is "yes" (or true), then the computer does something specific.



condition

If it's raining outside,



task

Take an umbrella.

Imagine you're deciding whether to carry an umbrella outside. The simple rule you might follow is: "If it's raining, then I'll take an umbrella. If it's not raining, I won't need one." This decision-making process is similar to how an "if statement" functions in programming. It acts like a rule or condition, essentially saying, "If a certain condition is true, then perform a specific action."

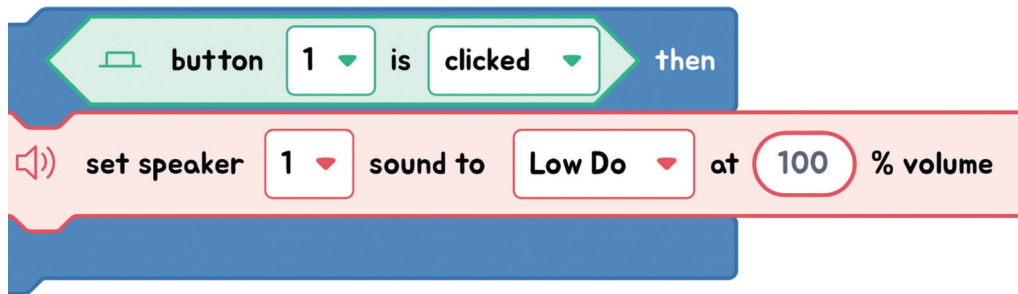
THINK ABOUT IT!

Think of another everyday situation where an "if" rule could apply. How would you phrase it?





EXPLANATION WITH CODE EXAMPLE



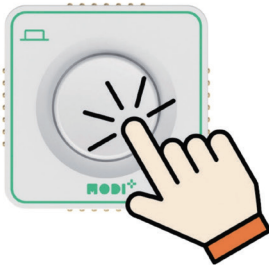
In block coding, we use the "if statement" to check conditions. If the condition is met (or true), the program will carry out a specific task. The "if statement" block visually separates the condition from the action with "if" and "then" parts. It examines the condition, and if the condition holds true, it proceeds to execute the task inside the block.



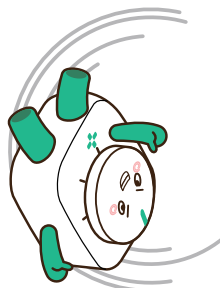
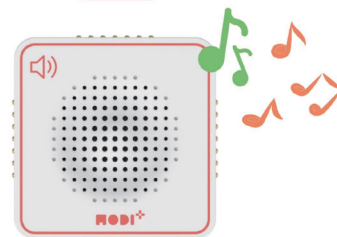
Code Insight

1. **Condition Checking:** The program starts by checking a specific condition, like seeing if a button has been pressed. So, the condition in the code is "If the button is clicked."
2. **Execution if True:** If the condition is met, then the tasks inside the "if" statement are carried out. These tasks specify the actions to take when the condition occurs. Thus, the task performed is "Set the speaker to play the 'low do' sound at 100% volume."

Condition



Task



Exercise

Finding Conditions and Tasks:

Look at the block example and identify the condition it checks and the task it performs!

Exercise 1

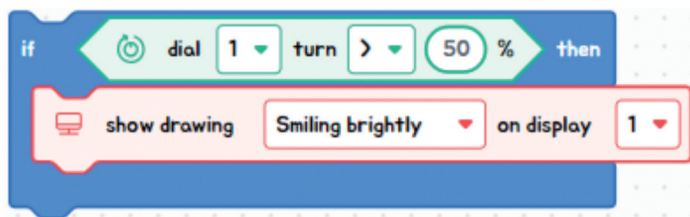


Condition(s)

Task(s)

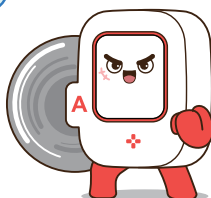


Exercise 2



Condition(s)

Task(s)

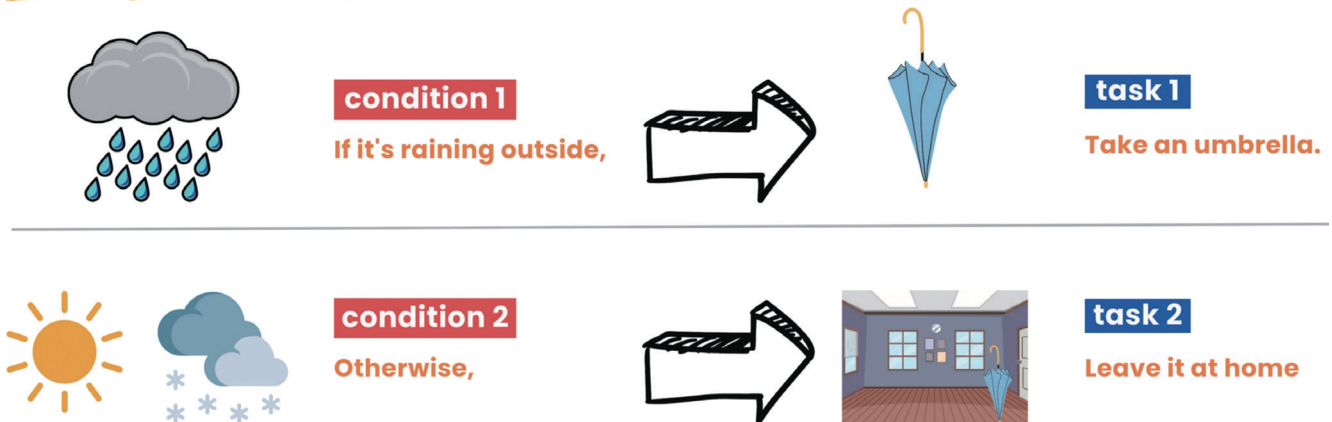




If else Statement

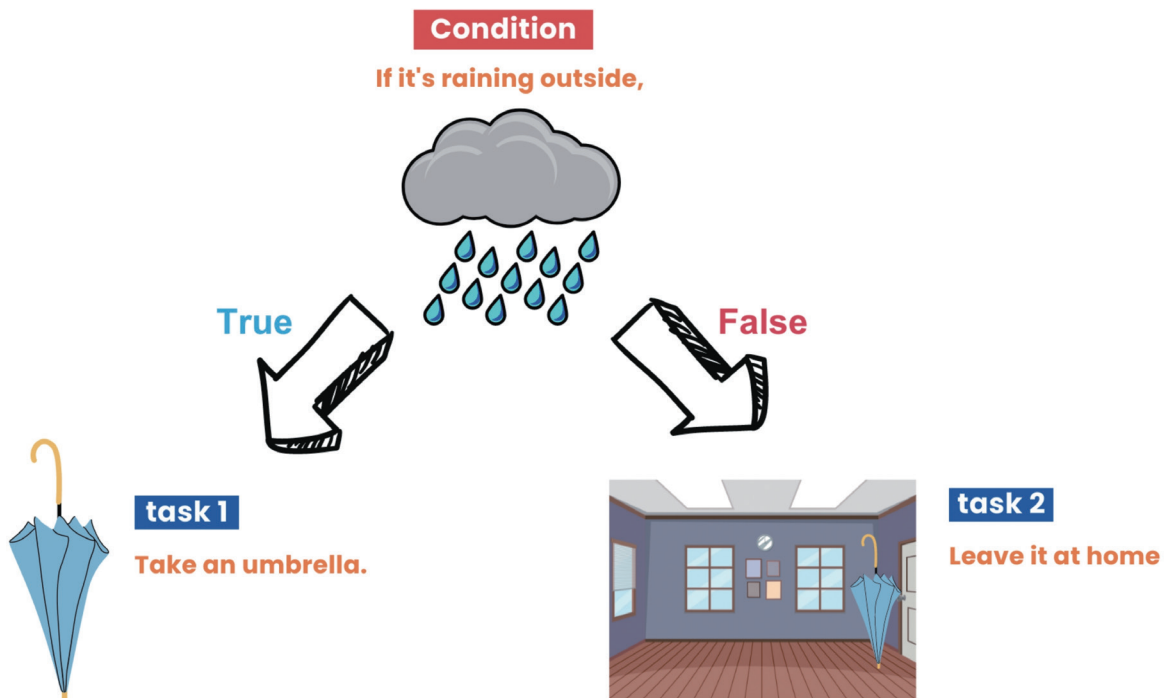
An 'If-Else' statement is like an 'If' statement with a backup plan. If the 'If' part isn't true, then the 'Else' part takes over and does something else instead.

✓ ELSE



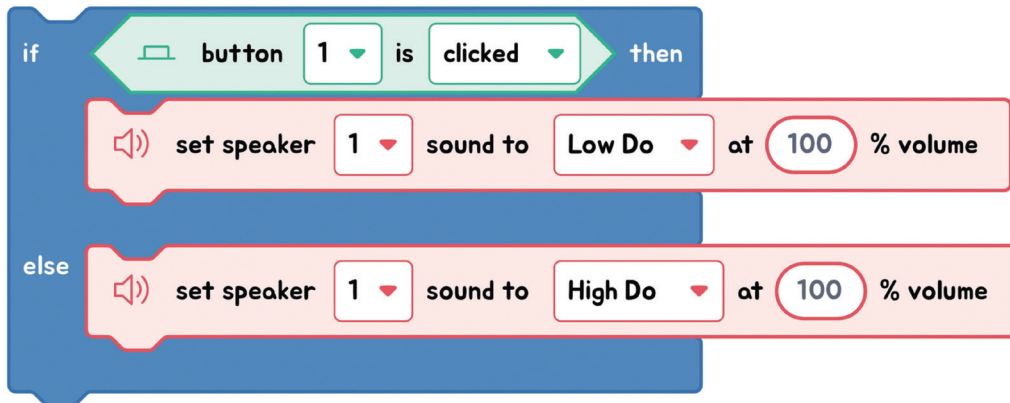
Imagine you're deciding whether to bring an umbrella. An 'If-Else' statement can help you with this!

If it's raining outside (this is the condition), you'll take an umbrella (this is the task). Otherwise (this is the Else part), you'll leave the umbrella at home. So, even though we only consider rain, Else takes care of everything else by default.

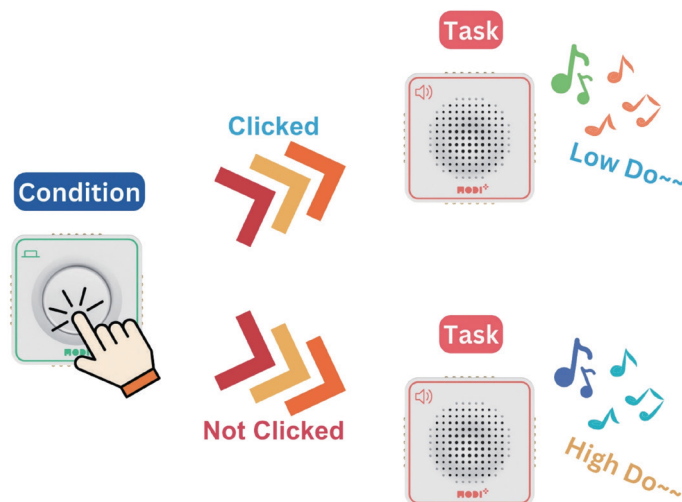


The key to 'If-Else' is that if the 'If' condition isn't met, the program will do what's in the 'Else' part instead.

EXPLANATION WITH CODE EXAMPLE



In block coding, the "If-Else" statement lets us choose between two actions based on a condition. If the condition is true, one action is taken; if not, a different action is executed. The "If-Else" block visually splits this choice, ensuring the program always has a clear direction, no matter the condition's outcome.



Code Insight

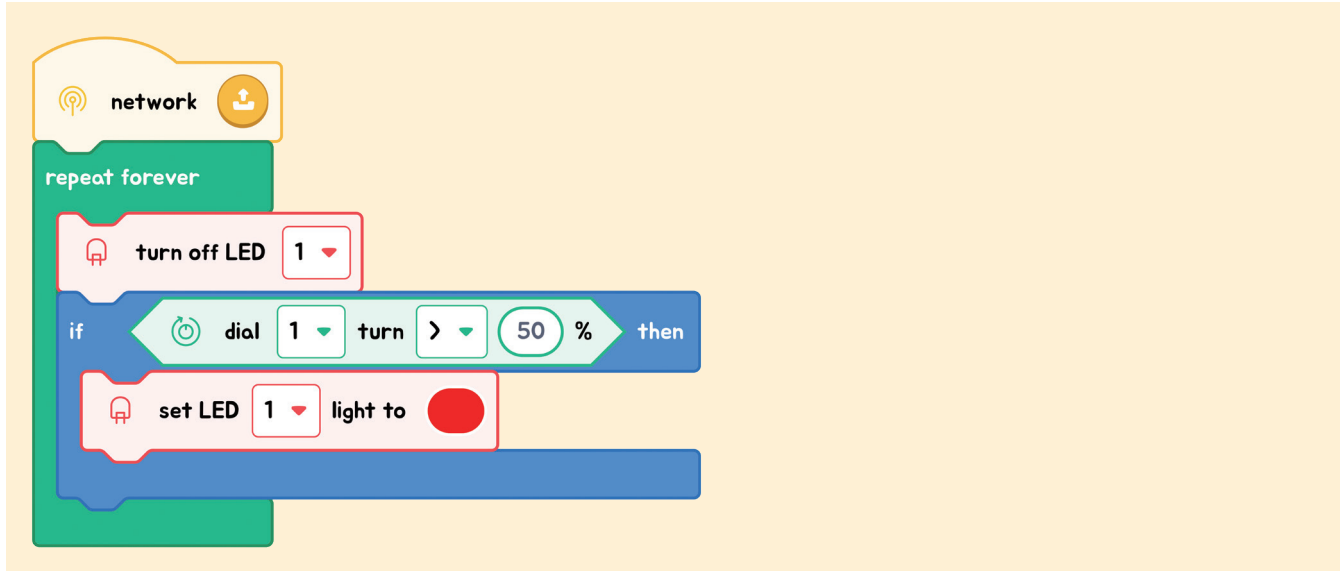
1. **Condition Checking:** The program starts by checking a specific condition, like seeing if a button has been pressed. So, the condition in the code is "If the button is clicked."
2. **Execution if True:** If the condition is met, then the instructions inside the "if" statement are carried out. These instructions specify the actions to take when the condition occurs. Thus, the task performed is "Set the speaker to play the 'low do' sound at 100% volume."
3. **Execution if False:** If the condition is not met, then the instructions inside the "else" statement are carried out. These instructions specify the actions to take when the condition is not met. Thus, the task performed is "Set the speaker to play the 'high do' sound at 100% volume."



Exercise

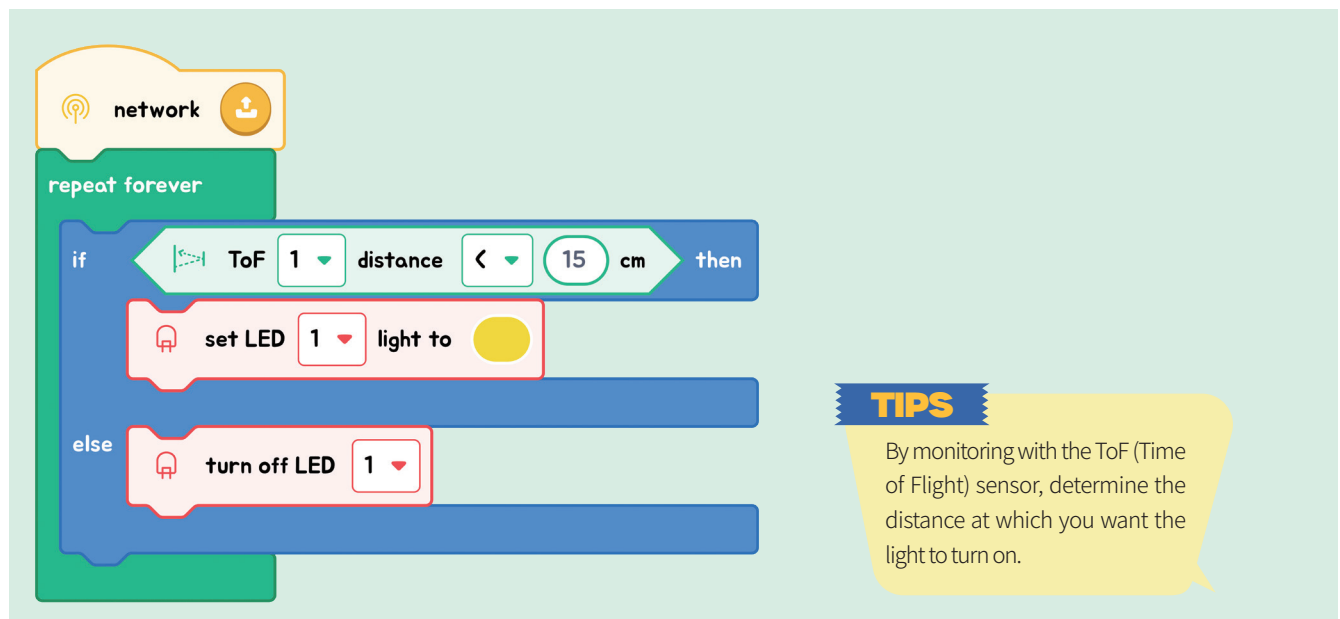
Exercise 1

Change this code to an 'If-Else' statement in the Code Editor.



Exercise 2

Use the Code Editor to create the following project: "We're making a sensor light. If an object is close to the sensor, the light will turn on. If there's no object nearby, the LED will be off."



TIPS

By monitoring with the ToF (Time of Flight) sensor, determine the distance at which you want the light to turn on.



Nested If Statement

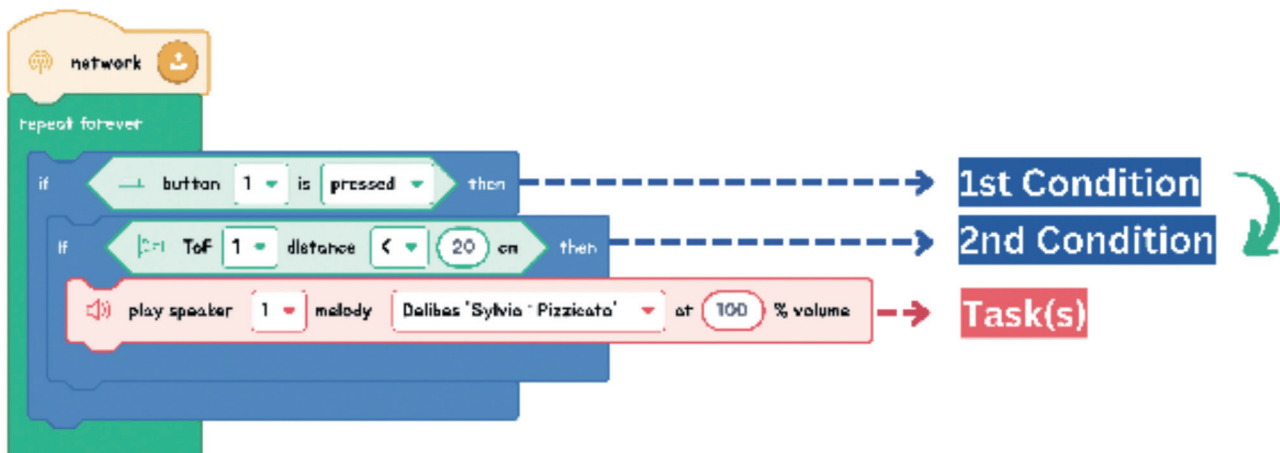
Nested if statements are like putting one decision inside another. First, you check a condition; if that's true, you check another one inside it. If this second condition is also true, then an action takes place. It's like solving a layered puzzle: solve one layer to get to the next!

✓ NESTED IF



First, you check if you're hungry (1st condition). If you are, you then look inside the fridge to see if there's food (2nd condition). If there's food available, you go ahead and make dinner (this is the task).

So, with a 'Nested If', you first decide if you're hungry, and only then do you consider whether there's food in the fridge to make a meal. Let's look at the example below!



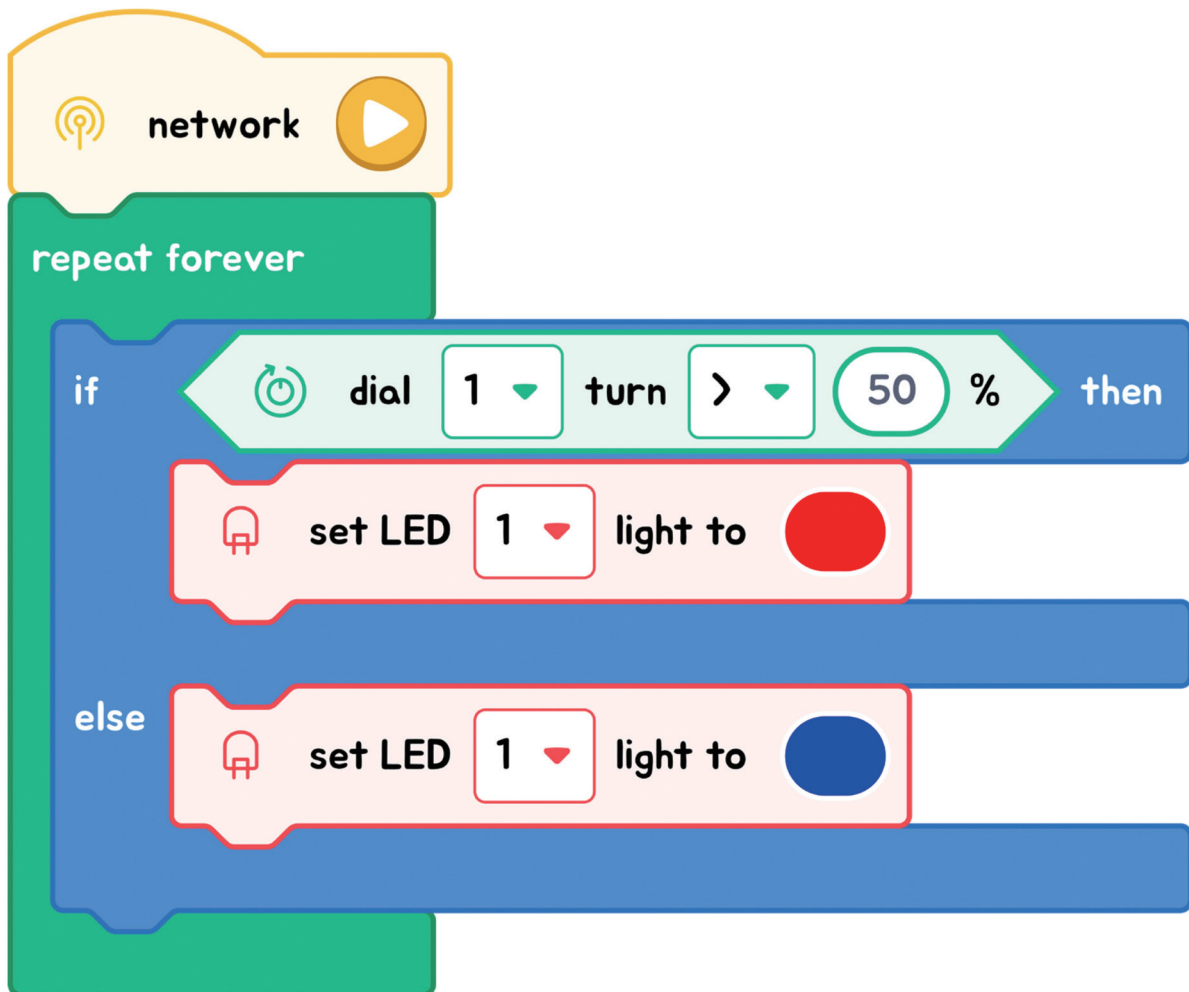
First, the program checks if a button is pressed (1st condition). If the button is indeed pressed, it then examines another condition: whether the distance measured by the ToF sensor is less than 20 cm (2nd condition). Only if both these conditions are met, does the program command the speaker to play the song.



Exercise

Exercise 1

Find the condition and task in the flowchart.



Your
Answer:

